

SAFAL TAMANG

Aspiring Data Scientist & AI/ML Engineer

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🌐 LinkedIn 🐙 GitHub

Professional Summary

Final-year BSc CSIT student specializing in AI engineering, Retrieval-Augmented Generation (RAG), and machine learning systems. Experienced in building custom RAG pipelines, scalable backend APIs with FastAPI, and vector search applications using Qdrant.

Technical Skills

Languages	Python, C (Basic), C++(Basic), SQL
Machine Learning	Scikit-learn, XGBoost, Random Forest, Gradient Boosting, Naive Bayes
Deep Learning	PyTorch
Generative AI & RAG	Retrieval-Augmented Generation (RAG), Qdrant Vector DB, Sentence-Transformers, Embeddings
Data Analysis & Viz	Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook
Backend Frameworks	FastAPI
Tools	Git, GitHub

Projects

Conversational RAG Document Assistant

Custom RAG & Multi-Turn Chat Backend

🐙 [GitHub Repository](#)

- Built a custom RAG backend with configurable document chunking, Sentence-Transformers embeddings, and Qdrant vector search, with Redis-based multi-turn memory.
- Developed modular FastAPI APIs for PDF/TXT ingestion, context retrieval, and LLM-powered QA, with automated booking extraction and SQLite persistence. ““

AgniDrishti Nepal

AI-Powered Wildfire Prediction Platform

🐙 [GitHub Repository](#) [Live Demo](#)

- Built an end-to-end predictive platform using Random Forest and Naive Bayes classifiers to compute real-time geospatial wildfire risk levels.
- Pipeline-integrated satellite and telemetry streams from NASA FIRMS, OpenWeatherMap, and Open-Elevation APIs for automated feature extraction.
- Deployed scalable FastAPI services backed by MongoDB for metadata persistence, secure JWT authentication, and historical trend dashboards.

Box Office Revenue Prediction

Machine Learning Web Application

🐙 [GitHub Repository](#)

- Developed ensemble regression algorithms including XGBoost, Gradient Boosting, and Stacking Regressors to forecast movie financial performance.
- Conducted extensive data cleaning, exploratory data analysis, and feature engineering with Pandas and NumPy, exposing inference pipelines via FastAPI.

Education

Madan Bhandari Memorial College

Kathmandu, Nepal

Bachelor of Science in Computer Science and Information Technology (BSc CSIT)

2023 – 2026 (Expected)

Dhulabari Mavi

Jhapa, Nepal

Higher Secondary Education (+2 Science)

Completed 2022

Certifications & Achievements

- **Second Runner-Up** – Tech Sprint Hackathon
- **DataCamp** – Data Scientist Career Track